Engineer guide: ingest a new dataset (A->Z)

Assumes you just cloned data-team. Goal: land data in landing_dev with two files (notebook + MD). Full dump by default; pilot only if you opt in.

Also mirrored in README.md.

A. Clone and open

```
git clone <repo>
# Open the data-team folder in Cursor
```

B. Create / activate `data-eng` venv

```
cd data-eng
python -m venv venv
.\venv\Scripts\Activate.ps1
```

Prefer this interpreter for notebooks when Cursor lets you pick it.

C. Install the ingestion package

```
pip install -e "./ingestion[dev]"
```

D. Configure `.env`

```
copy ingestion\.env.example ingestion\.env
```

Edit ingestion/.env:

- GCP_PROJECT, BQ_LOCATION=europe-west3
- INGEST_TARGET_DATASET=landing_dev
- GOOGLE_APPLICATION_CREDENTIALS = path to the shared SA JSON outside the repo
  (never under data-team/)
- Leave INGEST_IMPERSONATE_SA unset for key-only auth

E. Place the SA key outside the repo

Example:

C:\Users\<you>\.config\opentrace\ingest-dev.json

F. Check auth

```
cd ingestion
python -m ingestion check-auth
```

Expect kind: key_file and target: landing_dev.

G. Pick a full name

Pattern:

{organization}_{product_or_platform}_{dataset_full_name}

Example: world_bank_data360_harmonized_country_poverty

Use that same string for notebook, MD, and (later) package/table.

H. Copy an example as a template

Use:

- Notebook: world_bank_data360_harmonized_country_poverty_landing_dev.ipynb
- MD: world_bank_data360_harmonized_country_poverty.md
- Bootstrap: keep using _bootstrap.py (don't reinvent)

Rename copies to your full name, e.g.:

- .../<your_full_name>_landing_dev.ipynb
- .../<your_full_name>.md

I. Write the companion MD first (or in parallel)

Must include:

- What the source/dataset is and grain (what one row means)
- Column dictionary (business + _ metadata)
- How extract works (URL, paging, auth quirks)
- Full dump is default; how to set PILOT = True
- How to verify in BigQuery

J. Notebook setup cell

- Call _bootstrap so import ingestion works even on the wrong kernel
- get_settings(cli_target="landing_dev") + ensure_auth_safe
- TABLE / SOURCE_NAME = your full name
- PILOT = False by default

K. Extract (full by default)

- Pull the whole dataset (page if the API caps page size)
- Only if exploring: PILOT = True with a small filter - not the permanent default

L. Transform / normalize

- Flat columns suitable for landing
- Coerce types; keep stable identity keys for hashing

M. Add ingest metadata

Use shared helpers:

```
add_metadata(df, source=SOURCE_NAME, run_id=run_id, source_url=..., key_cols=[...])
```

N. Preview locally (optional)

Write parquet under ingestion/.out/ before BQ.

O. Load to BigQuery

```
load_dataframe(client, settings, ready, table=TABLE, mode="replace_partition", run_id=run_id)
```

Use the package load job path - not insert_rows_json.

P. Verify in BigQuery

- Table: landing_dev.<your_full_name>
- Row count ? source full count (if PILOT=False)
- Spot-check keys, nulls, sample values

Q. Review

Someone reviews notebook + MD + landing_dev table. Do not jump to prod yet.

R. (Later) Stage 2 - package

Only after review:

- Add package/ingestion/sources/<your_full_name>/ (source.yaml, extract.py, transform.py)
- Same full name; full dump default
- Small fixture + test
- Add to launch.json pick list

S. Package dry-run

```
cd data-eng\ingestion
$env:INGEST_OFFLINE_FIXTURES=1
python -m ingestion run <your_full_name> --dry-run
```

T. Package live to `landing_dev` (optional while validating CLI)

```
python -m ingestion run <your_full_name>
# or from data-eng: make ingest SOURCE=<your_full_name>
```

U. Prod `landing` only via CI

Needs INGEST_ALLOW_PROD=1 and CI=true. Do not set that for local day-to-day work.

V. What you should **not** do

- Put SA keys in the repo
- Start with package/CLI into landing
- Use short cryptic names as the primary identity
- Make a pilot slice the package default
- Hand-roll BQ streaming inserts for landing loads

W. Reference example

Working end-to-end pattern:

ingestion/notebooks/world_bank_data360_harmonized_country_poverty_*

X. Layout reminder

Path	Role
`data-eng/ingestion/notebooks/`	Stage 1 (your two files)
`data-eng/ingestion/package/`	Shared helpers + Stage 2 sources
`data-eng/ingestion/.env`	Local config (not committed secrets)

Y. Mental model

```
clone -> env + SA key -> notebook + MD -> landing_dev (full)
                              ? review
                    package source -> landing (CI)
```

Z. Done when

- Full (or intentional pilot) data is in landing_dev.<full_name>
- MD explains data + ingest concepts
- Review passed; Stage 2 only if/when promoting